Quality Control Pipeline — Holistic Overview

Disclaimer. All data are **synthetic / educational**. Automated metrics are **heuristic**. This is **not** clinical, diagnostic, or regulatory validation. Do not use outputs for real patient care.

Table of contents

1. Executive summary: what we prioritized
2. Why Prompt B (grounded) leads the clinical-facing app
3. End-to-end narrative: what we ran, in order
4. System architecture and data flow
5. Validation criteria table
6. How this differs from typical "LAB" Likert scales
7. Experimental design
8. Statistical analysis — deterministic batch (50 paired trials)
9. Statistical analysis — offline AI qualitative review
10. Interpreting deterministic vs narrative scores together
11. Files and roles
12. Technical details
13. Usage instructions

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1. Executive summary: what we prioritized

For an application framed around **high-risk behavioral health cohort review**, the highest-risk failure mode is **fabricated or misaligned numbers and structure** (wrong visit counts, invented totals, missing disclosures). Therefore we **prioritized clinical / analytic accuracy**—alignment with the **deterministic cohort table and retrieval payload**—over **perceived writing polish**.

- **Primary production report: Prompt B — Grounded Executive Report** (`qc/prompts/hw2_grounded_prompt.txt`): strict binding to structured counts, required headings, and verification language.
- **Contrast condition: Prompt A — Baseline** (`qc/prompts/hw2_baseline_prompt.txt`): encouraged qualitative prose and **explicitly does not require verbatim numeric binding**.

The **authoritative product-grade QC** for "is this report acceptable as grounded analytics?" is **deterministic validation + weighted validity score + strict pass**, computed in code (`qc/validators.py`, `qc/scoring.py`).

A **secondary, supplemental** layer runs an **AI reviewer offline** on saved report text only (`qc/run_ai_qualitative_review.py`): it scores **narrative** dimensions 1–5 **without** checking numeric truth (by design).

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2. Why Prompt B (grounded) leads the clinical-facing app

Concern	Prompt A (baseline)	Prompt B (grounded)
Numeric binding to cohort + JSON	Encourages approximate / qualitative description	**Hard rule:** counts and stated totals must match table/JSON; forbidden inventions
Required structure	Logical headers, flexible	**Exact** section order and labels (e.g., executive lines for visits / patients / lapsed follow-up)
Role in app	QC **contrast** prompt only	**`report_full` in pipeline = grounded output**; Shiny shows this as the main summary
Deterministic QC outcome (50-run batch)	**0%** strict pass rate (see §8)	**100%** strict pass rate

Decision: Even if a separate **narrative** reviewer might prefer looser prose on occasion, **Prompt B** is the appropriate default for anything described as **cohort-grounded executive reporting**, because it is the variant **engineered to satisfy the structural and numeric fidelity gates** that approximate "clinical analytic safety" in this educational stack.

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3. End-to-end narrative: what we ran, in order

Phase A — Deterministic paired experiment (50 trials)

1. Fixed **synthetic cohort** from `patients.db` (rule: **PHQ-9 > 15** and **safety_concerns = Y**).
2. **Same** deterministic **retrieval payload** and verification for every trial.
3. For each `trial_id` **0 ... 49**, the pipeline generated:
4. **Prompt A** completion → graded → one row in **`out/qc_batches/qc_50trials_20260505T130841Z.csv`** (`mode=baseline`).
5. **Prompt B** completion → graded → one row (`mode=grounded`).
6. **Total rows: 100** (50 baseline + 50 grounded).
7. **Statistical summary** for the batch is reflected in `out/qc_batches/qc_50trials_20260505T130841Z.md` (same numbers as in §8 below).

Phase B — Offline AI qualitative review (same 100 reports)

1. **Input:** archived CSV `out/qc_batches/qc_50trials_20260505T130841Z.csv`.
2. **Process:** each `report_text` sent to an OpenAI model with a fixed **narrative-only** reviewer prompt (see `qc/run_ai_qualitative_review.py`); **no numeric verification** in that step.
3. **Outputs:**
4. `out/ai_qualitative_review_results.csv` (row-level qualitative scores + JSON strengths/weaknesses).
5. `out/ai_qualitative_review_summary.md` (means, paired tests, bootstrap CIs, correlation with deterministic score).

Phase C — Application behavior (high level)

- **Live run:** builds **grounded report only** for the dashboard (`run_live_homework2_pipeline`), validates it live; **does not** overwrite batch QC artifacts (see `README_QC_VALIDATION.md`).

4. System architecture and data flow

This diagram matches the separation in `README_QC_VALIDATION.md`: **source of truth for deterministic grading** is the **retrieval payload + cohort truth**, not an LLM judgment.

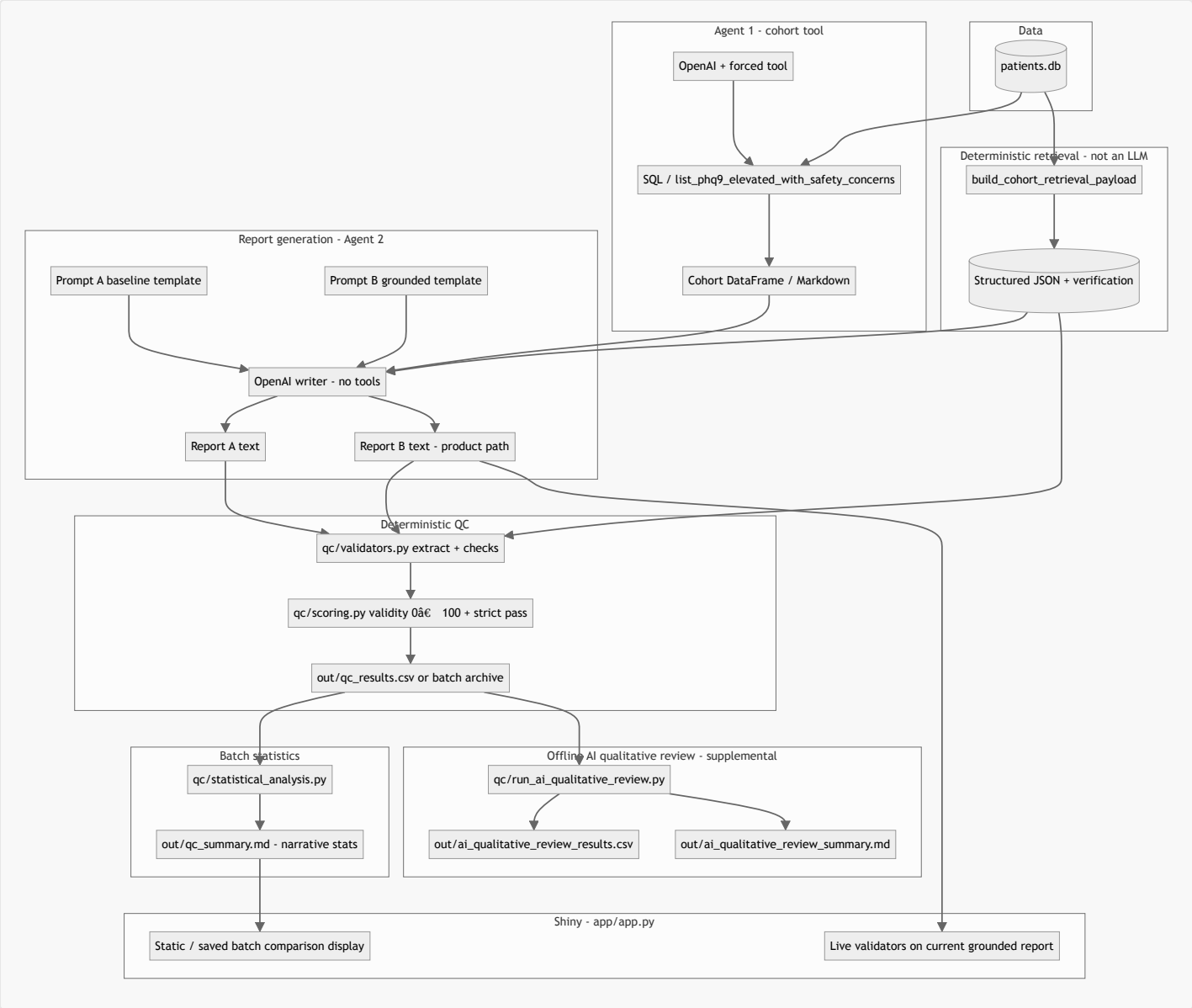

Figure — System architecture and data flow (§4).

Reading the diagram

- **Deterministic path:** cohort + payload → report text → **validators + scoring** → CSV → **Python statistics** (t-tests, McNemar, bootstrap, etc.).
- **AI qualitative path:** reads **already generated** report text; **does not** replace validators; **does not** change prompts or pipeline code when run.

5. Validation criteria table

A. Deterministic / structural (primary)

These are implemented in `qc/validators.py` and rolled up in `qc/scoring.py`. **Scale:** most sub-metrics are **rates in [0, 1]**; overall `validity_score_0_100` is a **weighted sum** (max 100 before penalty) minus a **capped hallucination-style penalty**. **Strict pass** is a **boolean gate** (see benchmarks below).

Dimension (representative)	Description (short)	Scale / method	Benchmark / gate (strict pass)
Numeric accuracy composite	Blend of visit / patient / lapsed fidelity and provider linkage	Rate **0–1** → contributes **weighted points** (weight **34**)	Part of `validity ≥ 80`; visit/patient/lapsed rates must equal **1.0** individually for pass
Patient count fidelity	Match of stated vs expected patient count	Rate **0–1** (weight **10**)	`patient_count_match_rate == 1.0` for pass
Provider visit fidelity	Alignment with retrieval provider visit structure	Rate **0–1** (weight **11**)	Implicit in weighted score under numeric/provider linkage
Lapsed follow-up fidelity	Match of lapsed follow-up reporting vs payload	Rate **0–1** (weight **9**)	`lapsed_followup_match_rate == 1.0` for pass
Required sections	Required `##` headings for **each prompt mode**	Rate **0–1** (weight **12**)	`required_sections_rate == 1.0` for pass
Retrieval check disclosure	Retriever / verification disclosures	Rate **0–1** (weight **8**)	In weighted score
Limitations disclosure	Stated limits of data / method	Rate **0–1** (weight **8**)	In weighted score
Medication theme mention	Presence of medication theme signal	Rate **0–1** (weight **8**)	In weighted score
Clinically unsupported numerals	Count of numerals that look like unsupported cohort claims	**Integer count**; penalty	**Must be 0** for strict pass
Unsupported patient identifiers	Citations not supported by payload	**Integer count**; penalty	**Must be 0** for pass
Unsupported provider strings	Provider tokens not supported	**Integer count**; penalty	**Must be 0** for pass
Validity score	Weighted rollup minus penalty + small concision term	**0–100**	`≥ 80` required for pass
Strict pass	Conservative "ship line" for this homework	**Boolean**	All of: validity ≥ 80, required sections 100%, patient/visit/lapsed match 100%, zero unsupported identifiers/numerals per `scoring.py`

Weights are defined in `WEIGHTS` in `qc/scoring.py`.

B. Offline AI qualitative (supplementary)

Source: reviewer prompt embedded in `qc/run_ai_qualitative_review.py`. Each dimension is scored **1–5 from report text alone**; the model is instructed **not** to verify numeric correctness.

Dimension	Description (from reviewer prompt)	Scale	Benchmark
Clarity	Organization, readability	**1–5** Likert-style	Descriptive only (no pass/fail)
Clinical usefulness	Actionability / insight for a care team	**1–5**	Descriptive
Coherence	Logical flow, contradictions	**1–5**	Descriptive
Completeness	Coverage of cohort patterns, provider/access, meds/docs, continuity	**1–5**	Descriptive
Overall quality	Holistic narrative quality	**1–5**	Descriptive; used vs `validity_score_0_100` in correlation table

Strengths / weaknesses are **lists of strings**; stored as JSON in the results CSV.

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6. How this differs from typical "LAB" Likert scales

Typical LAB / survey setups often ask human coders or students to rate outputs on **Likert scales** (e.g., 1–5 agreement with statements like "clear" or "accurate") **without** an external, machine-checkable **ground truth** for every item.

This project's primary QC is different:

1. **Ground-truth-anchored rates:** Many dimensions are **not** "opinion" — they are **computed against** the cohort table and retrieval JSON (counts, headings, disclosure heuristics).
2. **Single strict gate:** `passed_absolute_validity` collapses multiple **hard** requirements into one **audit-style** decision suitable for dashboards.
3. **0–100 validity** is a **weighted engineering score**, not a Likert average across human raters.
4. **The AI reviewer's 1–5 scales** *are* closer to **Likert-like narrative judgment**, but they are **explicitly secondary** and **explicitly barred** from adjudicating numeric truth (that stays in deterministic validators).

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7. Experimental design

Element	Specification
Prompts compared	**Prompt A:** `qc/prompts/hw2_baseline_prompt.txt` (qualitative baseline). **Prompt B:** `qc/prompts/hw2_grounded_prompt.txt` (grounded executive).
Pairing	Same `trial_id`, **same cohort**, **same retrieval payload** → **paired** comparisons.
Trials	**50** paired trials ⇒ **50** baseline + **50** grounded reports.
Sample size	**n = 50** paired differences per metric (deterministic validity and pass rate analyses use these pairs). Row-level deterministic output: **100** rows.

Element	Specification
Grader for experiment	**Deterministic codebase** (`validators` + `scoring`), identical rules for both modes aside from **mode-specific required headings**.
Supplement	Offline **AI qualitative** review on the **same 100** report texts (**separate** model call path).

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8. Statistical analysis — deterministic batch (50 paired trials)

Hypotheses (informal, for interpretation).
We expect **Prompt B** to produce **higher** deterministic validity scores and **higher** strict pass rates than **Prompt A**, because **B** is constrained to match structured truth.

Tests and estimands (implemented in `qc/statistical_analysis.py` ; narrative in batch markdown).

8.1 Outcome summaries (all rows pooled and by mode)

Quantity	Result (50-trial archived batch / `qc_50trials_*`)
Overall mean **validity_score_0_100**	**71.64**
Overall **95% CI** on mean validity (all 100 rows)	**[67.70, 75.58]**
Overall **pass rate** (strict)	**50.0%**
Prompt A mean validity	**51.64**
Prompt A pass rate + **Wilson 95%**	**0.0%** [CI **0.0–7.1**]
Prompt B mean validity	**91.64**
Prompt B pass rate + **Wilson 95%**	**100.0%** [CI **92.9–100.0**]
Structural error rate (reported aggregate)	**0.0%** for both modes in summary file

8.2 Pass-rate comparison (paired binary outcomes)

Quantity	Result
Pass-rate **difference** (Prompt B – Prompt A)	**+100.0** percentage points
Bootstrap 95% CI for that difference	**[100.0, 100.0]**
McNemar (statsmodels exact) **p-value**	**1.776e–15**
Discordant cells	baseline pass / grounded fail = **0**; baseline fail / grounded pass = **50**
Narrative conclusion in file	Pass-rate improvement statistically significant

8.3 Continuous paired score comparison (validity 0–100)

Quantity	Result
Paired *t*-statistic (grounded – baseline, paired by `trial_id`)	**95.37**
Cohen's *d* (paired formulation used in codebase)	**19.70**

Interpretation note: Cohen's ***d*** can be **very large** when paired lifts are **consistent** with **little variance**; it **does not** mean "multiplicatively better in a clinical sense."

8.4 Failure mode aggregation (frequency table across rows)

Failure mode	Count	Share
Did not pass absolute validity	**50**	**50.0%**
Visit count fidelity below 1.0	**50**	**50.0%**
Missing required sections	**50**	**50.0%**
Unsupported patient identifier citation(s)	**0**	**0.0%**
Unsupported provider name signal(s)	**0**	**0.0%**
Hallucinated state mention(s)	**0**	**0.0%**
Clinically unsupported numeral occurrence(s)	**0**	**0.0%**
Patient count fidelity below 1.0	**0**	**0.0%**
Lapsed follow-up count fidelity below 1.0	**0**	**0.0%**
Numeric accuracy below 0.60	**0**	**0.0%**
Confidence label match below 1.0	**0**	**0.0%**
Low/moderate confidence disclosure below 1.0	**0**	**0.0%**
Unsupported structured confidence claim(s)	**0**	**0.0%**
Grader stubs (unsupported_claims / confidence_misuse non-empty)	**0 / 0**	**0.0%**

(Counts can overlap across flags on the same rows.)

8.5 Confidence-validation block (included in markdown for framework compatibility)

- By-label summary for **neutral** dominance: validity **71.64**, numeric alignment **83.9%**, pass rate **50.0%**.
- **Pearson correlation** between upstream confidence columns and validity: **not computed** (insufficient variation in confidence scores).

Tests not used here: Analysis is **paired *t*-test style** statistics and **McNemar**, not multi-group ANOVA or regression, for this batch layer.

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9. Statistical analysis — offline AI qualitative review

Hypothesis (informal). Narrative reviewer scores **may diverge** from deterministic validity **because constructs differ** (readability vs grounding).

All numbers below match `out/ai_qualitative_review_summary.md` (model `gpt-4o-mini`, **n = 50** pairs).

9.1 Descriptive means (grounded – baseline)

Dimension	Baseline mean	Grounded mean	Difference (G – B)
clarity	**4.860**	**3.080**	**−1.780**

Dimension	Baseline mean	Grounded mean	Difference (G – B)
clinical_usefulness	**4.460**	**3.000**	**−1.460**
coherence	**4.940**	**3.380**	**−1.560**
completeness	**4.460**	**3.160**	**−1.300**
overall_quality	**4.460**	**3.040**	**−1.420**

9.2 Paired inference (paired difference per `trial_id`, two-sided *t* on mean difference vs 0)

Dimension	n	Mean diff	Paired *t*	*p*-value	Cohen's *d**	Bootstrap 95% CI (mean Δ)
clarity	**50**	**−1.780**	**−30.079**	**3.017e−33**	**−4.297**	**[−1.880, −1.660]**
clinical_usefulness	**50**	**−1.460**	**−20.506**	**1.078e−25**	**−2.929**	**[−1.600, −1.320]**
coherence	**50**	**−1.560**	**−18.040**	**2.788e−23**	**−2.577**	**[−1.720, −1.380]**
completeness	**50**	**−1.300**	**−14.212**	**5.231e−19**	**−2.030**	**[−1.480, −1.120]**
overall_quality	**50**	**−1.420**	**−18.665**	**6.471e−24**	**−2.667**	**[−1.560, −1.280]**

*Cohen's *d* for paired differences: **mean(Δ) / SD(Δ)** in script.

9.3 Relationship to deterministic QC (same enriched rows)

Measure	Result
Pearson ***r*** (`ai_overall_quality` vs `validity_score_0_100`)	**−0.888**
Mean **`ai_overall_quality`** | deterministic **passed**	**3.040** (*n = 50*)
Mean **`ai_overall_quality`** | deterministic **failed**	**4.460** (*n = 50*)

Interpretation: In this dataset, reports that **pass** deterministic strict checks tend to score **lower** on the subjective **overall narrative** Likert-like scale—consistent with **tighter, more templated prose** versus freer baseline narrative.

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10. Interpreting deterministic vs narrative scores together

Do not contradict the tables.

- **Deterministic QC:** Prompt B delivers **much higher fidelity—100% strict pass vs 0%**—with documented **Wilson intervals**, **McNemar**, and **paired validity lift**.
- **AI qualitative reviewer:** On **all five narrative dimensions**, **baseline mean scores were higher** than grounded (Δ **negative** everywhere); paired tests show **large negative mean differences** with **tight bootstrap intervals**.

Clinical product stance:
Prioritize Prompt B for anything described as **grounded, audit-style cohort reporting**. Treat the **offline AI reviewer** as **hypothesis generating** ("how does enforcing structure change perceived readability?"), **not** as a veto over deterministic grounding.

Suggested framing:

Tradeoff hypothesis: richer narrative flow (Prompt A) can read "smoother" to a reviewer that is **explicitly barred from grading numeric truth**, while stricter grounding (Prompt B) **buys deterministic validity at the expense of subjective narrative polish**—a trade acceptable for analytic integrity in this sandbox.

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11. Files and roles

Path	Purpose
`patients.db`	Synthetic SQLite visits/patients
`clinical_pipeline.py`	Agent 1 + retrieval + live grounded run path (no batch overwrite in app path)
`retrieval.py`	Deterministic cohort payload
`functions.py`	OpenAI client helpers, env loading
`dotenv_loader.py`	Loads `.env` from repo root + `HW2/.env`
`qc/validators.py`	All deterministic metrics
`qc/scoring.py`	`validity_score_0_100` + `passed_absolute_validity`
`qc/statistical_analysis.py`	Batch stats: Wilson, bootstrap pass difference, McNemar, paired *t*, Cohen's *d*, failure modes
`qc/report_generation.py`	Writes `qc_summary.md` narratives
`qc/run_hw2_qc_experiment.py`	Runs `_n` paired trials → writes **`out/qc_results.csv`** + summary + immutable archive under `out/qc_batches/`
`qc/run_ai_qualitative_review.py`	**Offline only:** AI reviewer on existing CSV → `ai_qualitative_review_*`
`qc/prompts/hw2_baseline_prompt.txt`	Prompt A template
`qc/prompts/hw2_grounded_prompt.txt`	Prompt B template
`app/app.py`	Shiny UI; live QC on current grounded report
`app/live_validation_card.py`	Live checklist presentation
`out/qc_batches/qc_50trials_*.csv`	**Immutable** archived 50-run row-level deterministic results
`out/ai_qualitative_review_results.csv`	Row-level AI qualitative scores
`out/ai_qualitative_review_summary.md`	AI qualitative statistical write-up
`README.md`	Project overview
`README_QC_VALIDATION.md`	Focused QC framework doc (unchanged companion)

Path	Purpose
`PUBLISH_CONNECT.md`	Posit Connect deployment notes

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12. Technical details

Topic	Detail
Python	See `.python-version`; **3.10–3.12** typical
Key packages	`pandas`, `numpy`, `openai`, `shiny`, `PyYAML`, `python-dotenv`, `statsmodels` (and **SciPy** transitively for some tests) — see `requirements.txt`
API keys	**`OPENAI_API_KEY`** required for any LLM step. Optional **`OPENAI_MODEL`**, **`HW2_AI_REVIEWER_MODEL`** for qualitative script. **Never commit** `.env`.
Database	Default `patients.db` next to `clinical_pipeline.py`; override with **`PATIENTS_DB`**
Batch QC env	**`HW2_QC_TRIALS`**, optional seeds via experiment script

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13. Usage instructions

Install

```
cd /path/to/DSAI-HW2/HW2
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Create **`.env`** at repo root or **`HW2/.env`** with at least:

```
OPENAI_API_KEY=sk-...
# optional:
# OPENAI_MODEL=gpt-4o-mini
# HW2_AI_REVIEWER_MODEL=gpt-4o-mini
```

Run the Shiny app (dashboard)

```
cd /path/to/DSAI-HW2/HW2
source .venv/bin/activate
shiny run app/app.py --reload
```

Run the 50-trial deterministic batch experiment (writes `out/qc_results.csv` + summary + archive)

```
cd /path/to/DSAI-HW2/HW2
source .venv/bin/activate
python qc/run_hw2_qc_experiment.py --n-trials 50
```

Long run on a laptop (macOS):

```
caffeinate -dimsu python qc/run_hw2_qc_experiment.py --n-trials 50
```

Run the offline AI qualitative review on an existing 100-row batch CSV

```
cd /path/to/DSAI-HW2/HW2
source .venv/bin/activate

python qc/run_ai_qualitative_review.py \
  --input out/qc_batches/qc_50trials_20260505T130841Z.csv \
  --output out/ai_qualitative_review_results.csv \
  --summary out/ai_qualitative_review_summary.md
```

Resume after interruption (skips rows with `ai_review_available` already true):

```
python qc/run_ai_qualitative_review.py \
  --input out/qc_batches/qc_50trials_20260505T130841Z.csv \
  --output out/ai_qualitative_review_results.csv \
  --summary out/ai_qualitative_review_summary.md \
  --resume
```

Regenerate `ai_qualitative_review_summary.md` from an existing completed results CSV:

```
python qc/run_ai_qualitative_review.py \
  ...same arguments... \
  --resume
```

Where to read results

Artifact	Content
`out/qc_batches/qc_50trials_*.md`	Deterministic statistical narrative for that batch
`out/qc_batches/qc_50trials_*.csv`	Raw per-report deterministic metrics
`out/ai_qualitative_review_summary.md`	**All** qualitative statistical tests

Cross-reference

For a shorter QC-only checklist and file chart, continue to `README_QC_VALIDATION.md`. This document (`README_QUALITY_CONTROL_PIPELINE.md`) is the **holistic** narrative with **full statistical tables** and the **clinical-accuracy-vs-narrative** framing.